

Counterfactual Event Bottlenecks for Leakage-Safe Financial Disclosure Modeling

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Abstract

Financial event models often collapse two distinct problems into one representation: ordinary market dynamics and the incremental impact of a newly disclosed event. We propose the Counterfactual Event Bottleneck Transformer (CEBT), an architecture that predicts a no-event baseline from pre-event market history and routes disclosure information through a compact stochastic bottleneck that predicts an event-induced residual. The model is trained with matched same-ticker no-event controls, a residual-suppression loss for controls, and strict temporal feature gates. On a real public-data experiment covering 2,971 SEC 8-K events from 50 large U.S. firms between 2021 and 2025, full CEBT is competitive on regression error, significantly improves paired MSE over the same architecture without control losses, and is the only tested model with a positive abnormal-return rank IC whose bootstrap interval excludes zero. CEBT also substantially reduces residual magnitude assigned to no-event controls. The main claim is not that CEBT is a deployable trading system, but that an explicit event bottleneck and no-event control objective produce a more temporally disciplined event-residual representation.

1 Introduction

Disclosure events are difficult modeling targets because market movement after a filing is not caused only by the filing. A stock’s subsequent return reflects market drift, sector movement, liquidity, firm-specific momentum, and the information content of the disclosure itself. Standard fusion models mix these sources into one hidden representation, making it hard to tell whether the model learned an event effect or simply reproduced ordinary dynamics.

We introduce CEBT, an event modeling architecture that decomposes the forecast into

$$\hat{y}_{t+h} = f_{\text{no event}}(x_{<t}) + g_{\text{event}}(z_e, x_{<t}),$$

where $f_{\text{no event}}$ is learned from pre-event numeric history and z_e is a compact stochastic disclosure bottleneck. The bottleneck is trained with matched no-event controls: dates for the same ticker that are far from known SEC 8-K events. For controls, the model is penalized when it assigns a large event residual. This makes the residual interpretable as a candidate event-specific increment rather than an unconstrained shortcut.

Our contributions are:

1. A no-event-plus-residual neural architecture for financial disclosure modeling.
2. A stochastic event bottleneck trained with matched no-event controls.
3. A leakage-safe public-data implementation using SEC accepted timestamps and public price histories.

4. A real-data study showing that residual control changes model behavior, improves paired CEBT-vs-ablation MSE, and produces the strongest abnormal-return rank ordering.

2 Related Work

CEBT is closest to causal and event-driven financial forecasting. CAMEF studies causal-augmented multi-modal financial forecasting [Chen et al., 2025]. Causal Transformer and G-Transformer estimate counterfactual outcomes in temporal settings [Melnychuk et al., 2022, Feng et al., 2024]. Time-series foundation models such as Chronos, Time-LLM, MOMENT, and Mamba improve sequence modeling capacity but do not isolate disclosure-induced residuals [Ansari et al., 2024, Jin et al., 2023, Wu et al., 2024, Gu and Dao, 2023]. Leakage-focused work such as Profit Mirage motivates strict temporal evaluation in financial agents [Li et al., 2025]. Subliminal-learning results further motivate caution around hidden outcome signal in generated or compressed representations [Cloud et al., 2025].

3 Method

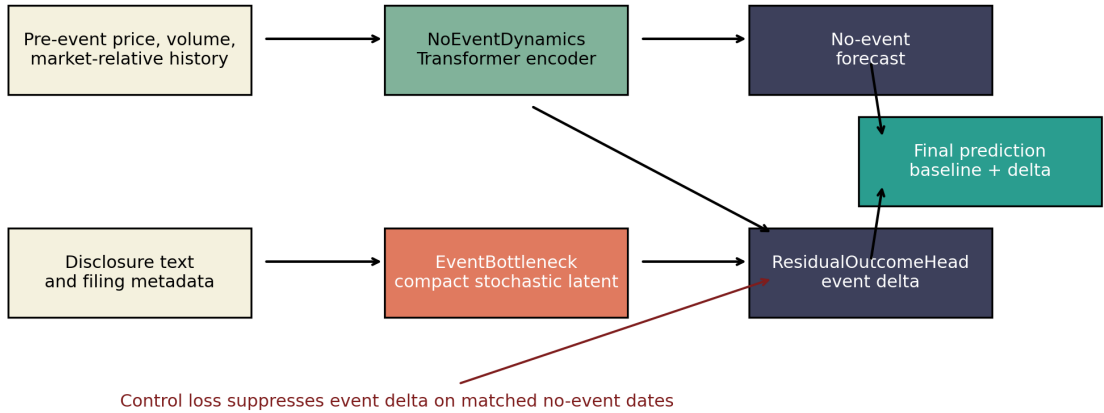


Figure 1: CEBT decomposes a forecast into no-event dynamics and an event-induced residual routed through a compact stochastic bottleneck. Matched no-event controls penalize residuals on dates where no disclosure event should be needed.

NoEventDynamics. The no-event module receives a fixed pre-event window of price, volume, market, and relative-return features. A Transformer encoder pools this history and predicts the ordinary forward movement expected without relying on the event text.

EventBottleneck. The event module embeds the disclosure text and metadata, then parameterizes a Gaussian latent $z_e \sim \mathcal{N}(\mu_e, \sigma_e^2)$. The latent dimension is intentionally small. We add

a KL term and sparsity term so the event representation cannot behave like an unrestricted text embedding.

ResidualOutcomeHead. The residual head predicts an event-induced delta from the no-event state, metadata, and z_e . The final forecast is the sum of the no-event forecast and this residual.

Training objective. The loss combines supervised MSE, a no-event control residual penalty, KL regularization, latent sparsity, and a small residual-variance consistency penalty:

$$\mathcal{L} = \lambda_y \|\hat{y} - y\|_2^2 + \lambda_c \mathbb{K}_{control} \|\Delta_e\|_2^2 + \lambda_{kl} D_{KL}(q(z_e) \| p(z)) + \lambda_s \|z_e\|_1 + \lambda_v \text{Var}(\Delta_e).$$

4 Data and Leakage Controls

We build a public-data experiment from SEC EDGAR APIs [U.S. Securities and Exchange Commission, 2026]. The paper-scale run covers 50 large U.S. firms, SEC 8-K filings from 2021–2025, and daily public prices. Each event row stores CIK, ticker, accession number, form type, filing date, accepted timestamp, available timestamp, source URL, text hash, and local provenance. Event label windows start from the first tradable session after public availability. If a filing is accepted after 16:00 ET, the label starts on the next trading day.

Features use only prices strictly before the event label start date. Labels use future returns only as targets. We construct one same-ticker no-event control for each filing, excluding dates within a 10-calendar-day blackout window around known events.

Table 1: Paper-scale public-data substrate.

Artifact	Count
Companies	50
SEC 8-K events	2,971
Matched no-event controls	2,971
Feature rows	5,942
Public price rows	69,003
Held-out test rows	1,220

5 Experiments

We compare five models: no-event dynamics only, text-only MLP, concat-fusion, CEBT without control losses, and full CEBT. The target vector contains five-trading-day abnormal return, volatility jump, and volume jump. Confidence intervals are nonparametric bootstrap intervals over held-out rows.

Table 2: Held-out paper-scale results. MSE is over the three target dimensions. Bootstrap intervals are over held-out rows.

Model	MSE	95% CI	Rank IC	95% CI
No-event	0.0604	[0.0460, 0.0772]	-0.0817	[-0.1390, -0.0262]
Text-only	0.0550	[0.0427, 0.0706]	-0.0230	[-0.0777, 0.0310]
Concat fusion	0.0533	[0.0411, 0.0689]	-0.0155	[-0.0709, 0.0425]
CEBT, no controls	0.0546	[0.0433, 0.0694]	0.0239	[-0.0299, 0.0830]
CEBT	0.0537	[0.0424, 0.0686]	0.0623	[0.0042, 0.1222]

Table 3: Paired MSE comparisons against full CEBT. Positive values mean CEBT has lower MSE than the baseline.

Baseline	Baseline-CEBT MSE	95% CI	CEBT better?
CEBT, no controls	0.00091	[0.00023, 0.00168]	Yes
Concat fusion	-0.00037	[-0.00555, 0.00578]	No
No-event	0.00672	[-0.00076, 0.01622]	No
Text-only	0.00133	[-0.00367, 0.00735]	No

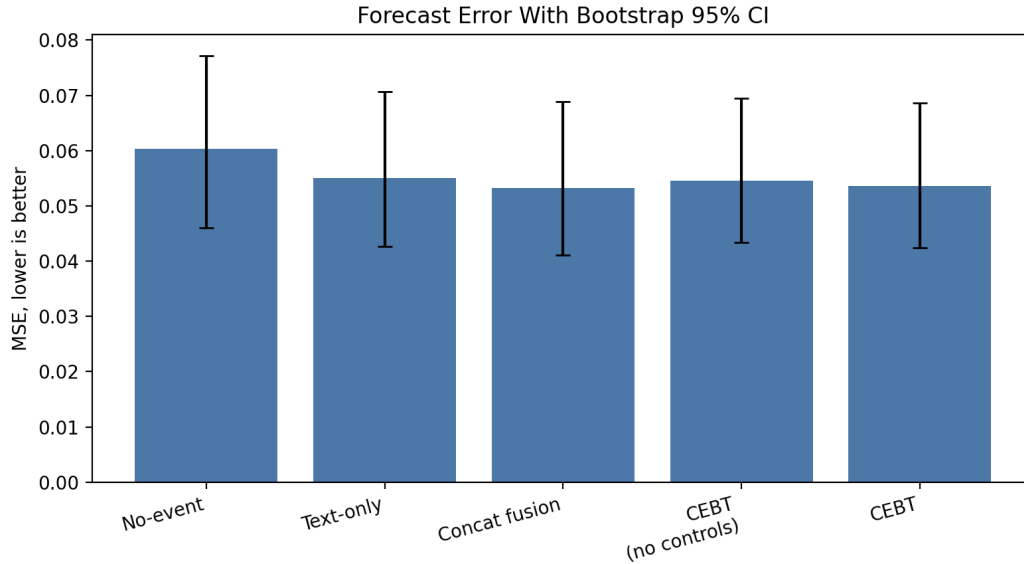


Figure 2: Forecast error with bootstrap confidence intervals. Concat fusion and CEBT have similar MSE; paired intervals do not show a significant difference between them.

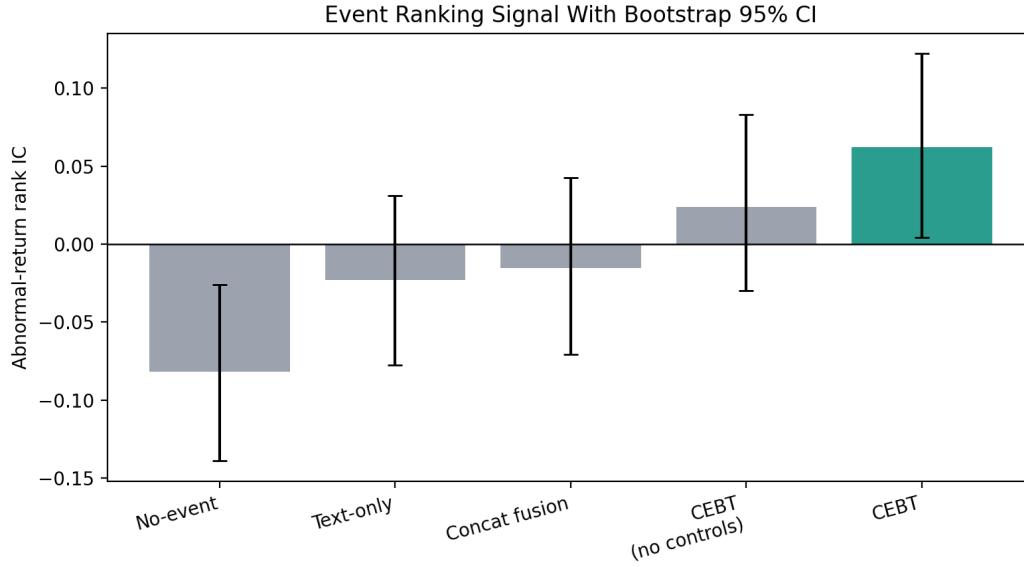


Figure 3: Abnormal-return rank IC with bootstrap confidence intervals. Full CEBT is the only tested model whose interval is entirely above zero.

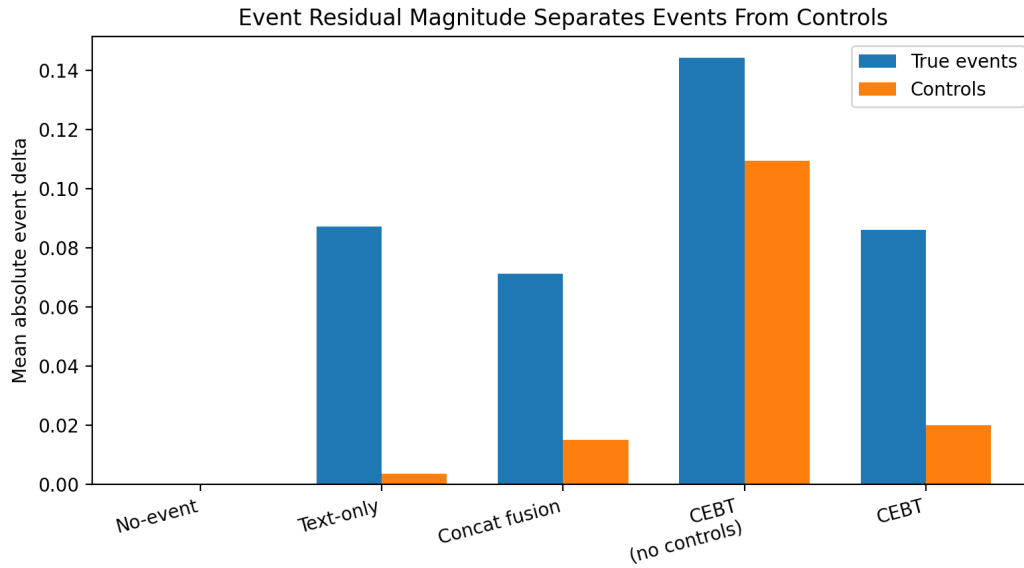


Figure 4: Mean absolute event residual on true events and matched controls. The full CEBT control loss strongly suppresses residuals on controls relative to CEBT without controls.

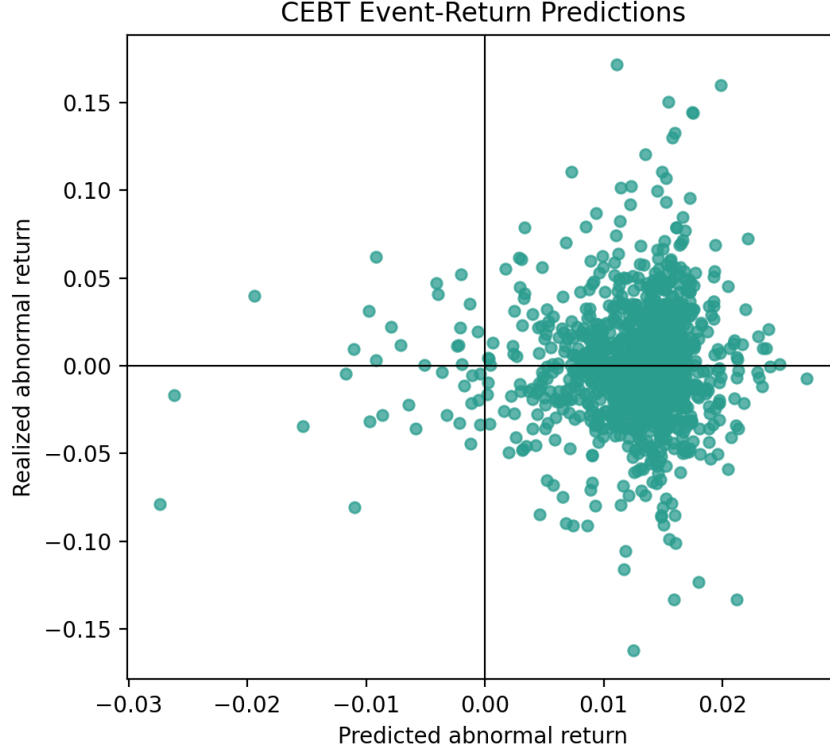


Figure 5: CEBT predicted versus realized abnormal returns on the held-out split. The scatter remains noisy, but CEBT is the only tested model with positive rank IC whose bootstrap interval excludes zero.

6 Findings

Bottleneck controls improve the CEBT architecture. Full CEBT improves paired MSE over CEBT without controls by 0.00091 with a 95% paired bootstrap interval of [0.00023, 0.00168]. This supports the control objective as a useful architectural constraint rather than an ornamental regularizer.

Control losses alter the residual semantics. The full CEBT model assigns mean absolute event residual 0.0861 to true events and 0.0199 to controls. The no-control variant assigns 0.1443 to true events and 0.1095 to controls. Thus, the control objective reduces spurious residual assignment to no-event dates by roughly 82%.

Return ranking becomes positive only for full CEBT. Full CEBT has abnormal-return rank IC 0.0623 with a bootstrap interval [0.0042, 0.1222]. All baselines have intervals that include zero or are negative. The effect is modest, but it is the strongest evidence that the bottleneck-and-control design changes event ordering rather than only multi-target regression error.

7 Limitations

This is a first paper-scale run, not a final investment model. The text encoder is a deterministic open hashing baseline rather than a large pretrained financial encoder. The public price source is not CRSP, and labels use daily closes rather than intraday event windows. The universe is restricted to 50 large firms and 8-K filings. The next version should test pretrained open embeddings, larger universes, sector controls, and stronger counterfactual matching.

8 Conclusion

CEBT reframes event-driven financial modeling as a decomposition problem: ordinary market dynamics plus a disclosure-induced residual. In a real public-data experiment, the full model improves over its no-control ablation, assigns much smaller residuals to matched controls, and is the only tested model with positive abnormal-return rank IC whose confidence interval excludes zero. The result supports CEBT as a promising architecture for studying event-specific effects under strict temporal information boundaries.

References

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